
PROJECT REPORT

Titanic Survival Prediction using Logistic Regression

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1. Introduction

The sinking of the RMS Titanic on 15 April 1912 remains one of the most widely studied disasters in maritime history, and the passenger manifest data associated with it has become a standard benchmark dataset for binary classification in machine learning. This report documents the development of a Logistic Regression model that predicts whether a given Titanic passenger survived the disaster, using demographic and travel-related attributes such as sex, age, passenger class, family size aboard, fare paid, and port of embarkation.

Beyond training a classifier, this report walks through the complete applied machine learning pipeline expected of a working Data Scientist: understanding the raw data, exploring it visually and statistically, cleaning and encoding it correctly, engineering a model-ready feature set, training and evaluating a Logistic Regression classifier, tuning its hyperparameters, and translating the resulting model into business-relevant insight for a maritime safety context.

2. Problem Statement

Acting as a Data Scientist for a maritime safety organization, the objective is to build a predictive model that estimates the probability of passenger survival based on demographic and travel-related information, and to use that model to surface the factors most strongly associated with survival. The target variable is binary:

- Survived = 0 -> Passenger did not survive
- Survived = 1 -> Passenger survived

Because the outcome is binary and several candidate predictors (sex, class, age, fare, family size) are hypothesized to influence it, Logistic Regression is a natural, interpretable first choice of model - its coefficients directly quantify each feature's association with survival odds.

3. Dataset Description

The dataset was supplied as `train_and_test2.csv`, sourced from the Kaggle "Titanic" dataset (heptapod). The raw file contains 1309 rows and 28 columns.

3.1 Data Cleaning Findings

- 19 columns (named `zero`, `zero.1`, ... `zero.18`) were identified as encoding artifacts: every value in these columns is 0, so they carry no predictive information. They were dropped.
- The target column was named `"2urvived"` in the raw file and was renamed to `"Survived"` for clarity.
- Critical data-quality finding: rows 892-1309 (418 rows) - corresponding to Kaggle's original `test.csv` split, whose true labels were never publicly released - all carry a placeholder `Survived = 0` value. These are not genuine labels. They were excluded, leaving 891 rows with authentic ground truth for modeling. Leaving these rows in silently biases the model toward predicting non-survival and would have understated true model performance.
- Sex, Pclass, and Embarked are already numerically encoded in this file version (Sex: 0/1, Pclass: 1/2/3, Embarked: 0/1/2), unlike the original raw Titanic CSV which uses text labels.

- Only Embarked has missing values (2 rows); Age and Fare have no missing values in this pre-processed file.

3.2 Feature Summary

Feature	Type	Description
Passengerid	Numerical (ID)	Unique passenger identifier
Age	Numerical	Passenger age in years
Fare	Numerical	Ticket fare paid
Sex	Categorical	0 = Male, 1 = Female
SibSp	Numerical	Number of siblings/spouses aboard
Parch	Numerical	Number of parents/children aboard
Pclass	Categorical	Ticket class: 1, 2, or 3
Embarked	Categorical	Port of embarkation (coded 0/1/2)
Survived	Target	0 = Did not survive, 1 = Survived

4. Exploratory Data Analysis

After removing the placeholder rows, the modeling dataset contains 891 passengers. The overall survival rate is 0.384 - meaning the classes are moderately imbalanced (roughly 62% did not survive vs. 38% survived), not extreme enough to require resampling but worth keeping in mind when interpreting accuracy.

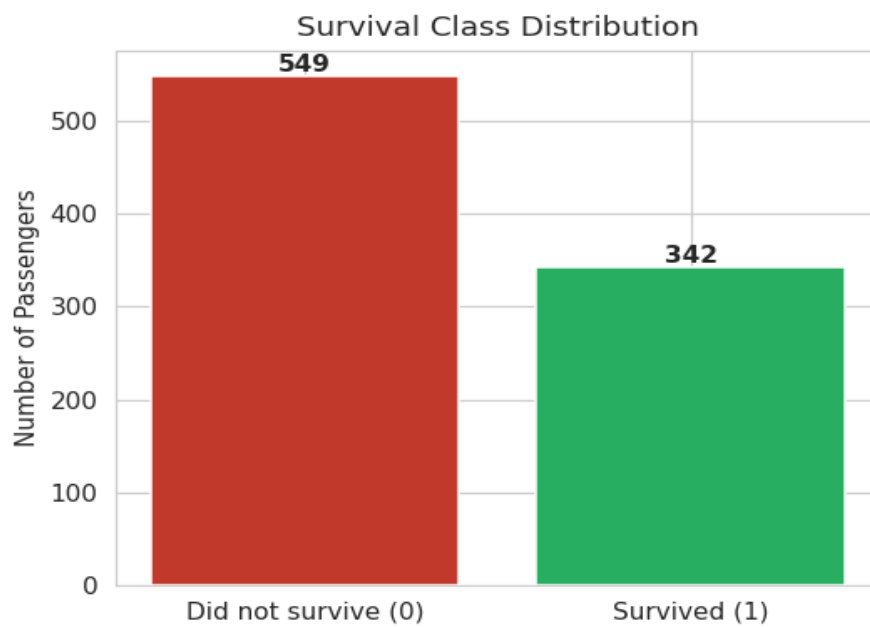


Figure 1. Survival class distribution.

4.1 Correlation Analysis

The correlation heatmap below shows Sex as the feature most strongly correlated with Survived (positive), and Pclass as strongly negatively correlated (higher class number = lower survival).

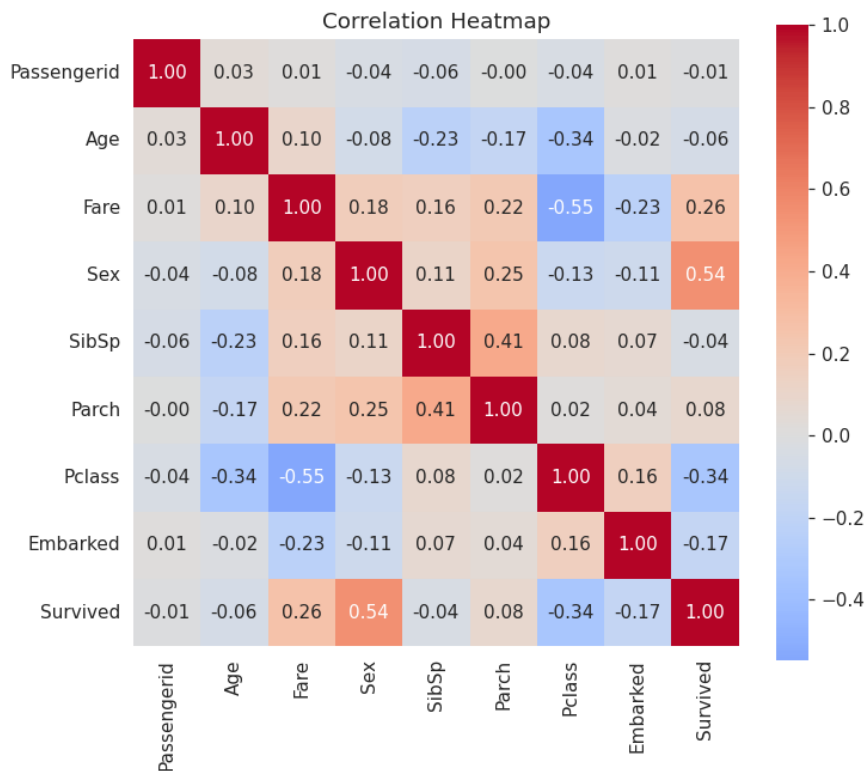


Figure 2. Correlation heatmap of numeric/encoded features.

4.2 Distributions and Outliers

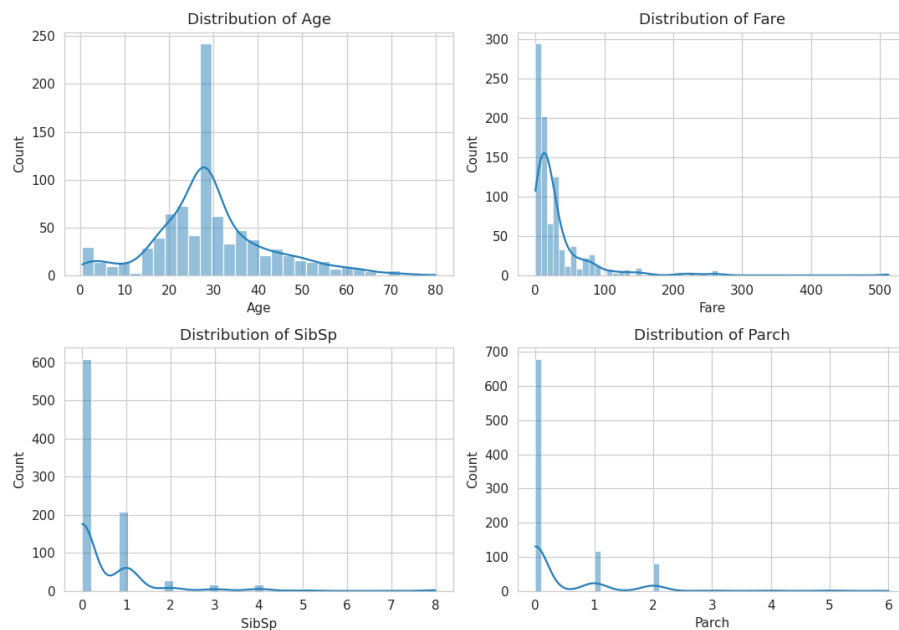


Figure 3. Histograms of numerical features.

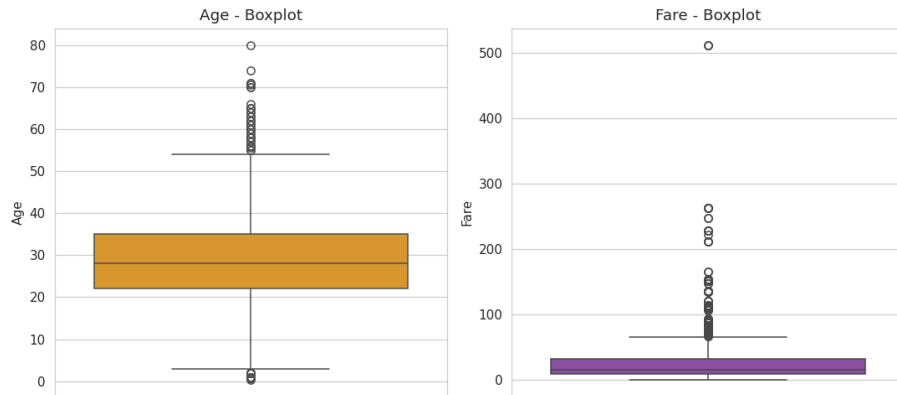


Figure 4. Boxplots of Age and Fare showing outliers.

Using the IQR method, Age shows 66 outliers and Fare shows 116 outliers - the latter reflecting a small number of passengers who paid very high fares (first-class, premium cabins).

4.3 Survival by Category

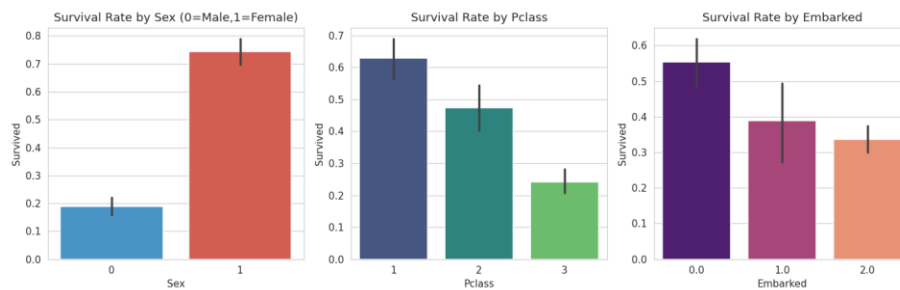


Figure 5. Survival rate by Sex, Pclass, and Embarked.

Survival rate by Sex: Male approximately 0.189, Female approximately 0.742. Survival rate by Pclass: 1st approximately 0.63, 2nd approximately 0.473, 3rd approximately 0.242.

4.4 Pairplot

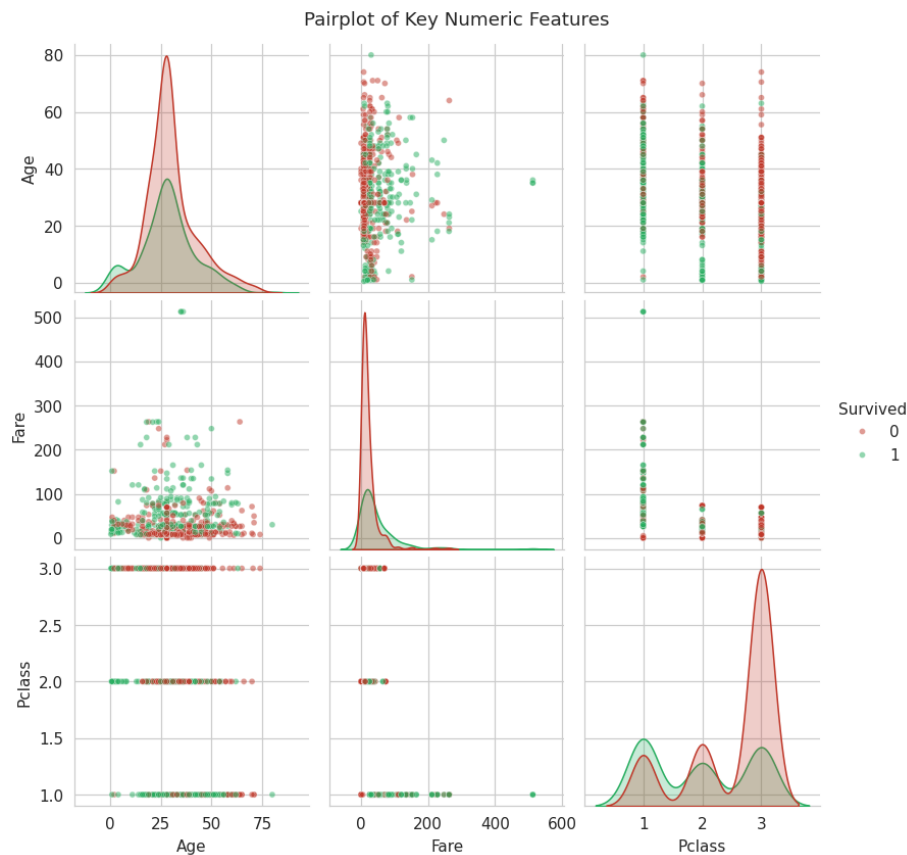


Figure 6. Pairwise relationships between Age, Fare, and Pclass, colored by survival.

4.5 EDA Observations

- Sex is the feature most highly correlated with survival.
- The dataset is moderately imbalanced (approximately 62/38 split), which is factored into metric choice later.
- Sex, Pclass, and Fare are the most visually and statistically useful predictors; Age and family-size features (SibSp, Parch) add smaller but non-trivial signal.

5. Data Preprocessing

Four preprocessing steps were applied before model training, each necessary for a different reason:

- Missing value handling - the 2 missing Embarked values were imputed with the column's mode (most frequent port). Logistic Regression cannot fit rows containing NaNs, and mode imputation is a safe, low-bias choice for a categorical feature missing in under 0.5% of rows.
- Encoding - Pclass and Embarked were one-hot encoded (drop_first=True) because they are nominal categories; representing them as raw integers would incorrectly imply an ordinal/linear relationship (e.g. that class 3 is 'three times' class 1). Sex was already binary and required no further encoding.

- Feature scaling - all features were standardized (zero mean, unit variance) with StandardScaler, since Logistic Regression's gradient-based solver and L1/L2 regularization are sensitive to feature magnitude; without scaling, high-magnitude features like Fare would dominate the penalty term.
- Train-test split - an 80/20 stratified split (train: 712 rows, test: 179 rows) was used to obtain an honest estimate of generalization performance, with stratification preserving the approximately 62/38 survival ratio in both sets.

6. Model Building

A Logistic Regression classifier (scikit-learn, max_iter=1000) was trained on the scaled training features.

6.1 Model Coefficients

Feature	Coefficient
Sex	1.2653
Pclass_3	-1.0707
Age	-0.4939
Pclass_2	-0.3744
SibSp	-0.261
Embarked_2.0	-0.187
Fare	0.1239
Embarked_1.0	0.0794
Parch	-0.0696

Intercept: -0.6588

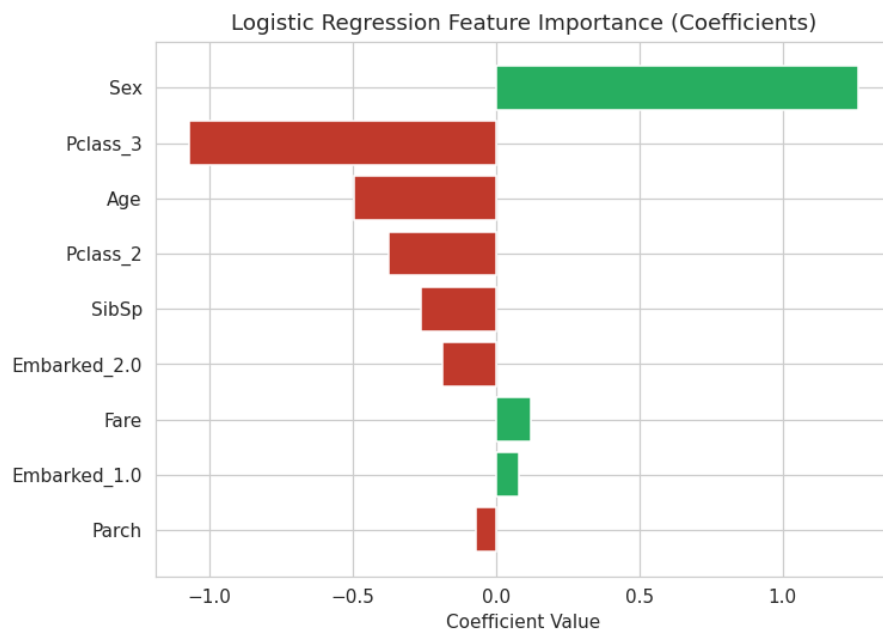


Figure 7. Feature importance ranked by absolute coefficient magnitude.

Sex carries by far the largest positive coefficient, meaning - holding other features constant - being female is associated with substantially higher odds of survival. Pclass_3 (third class) carries the largest negative coefficient, meaning third-class passengers had substantially lower survival odds than first-class passengers.

7. Model Evaluation

Metric	Value
Accuracy	0.8045
Precision	0.7931
Recall	0.6667
F1 Score	0.7244
ROC-AUC	0.8423

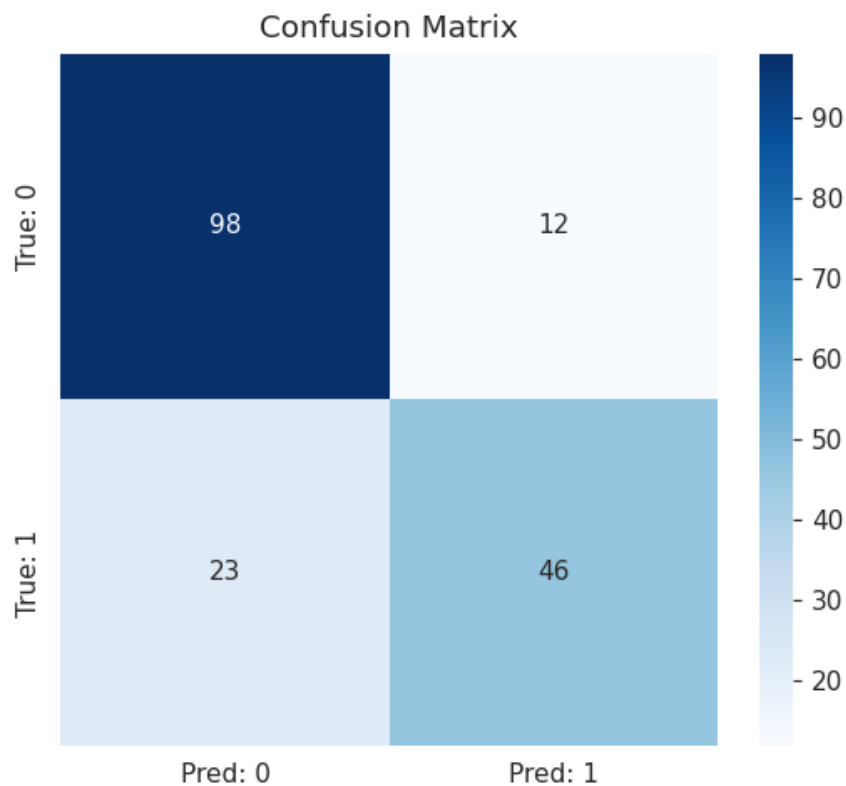


Figure 8. Confusion matrix on the held-out test set.

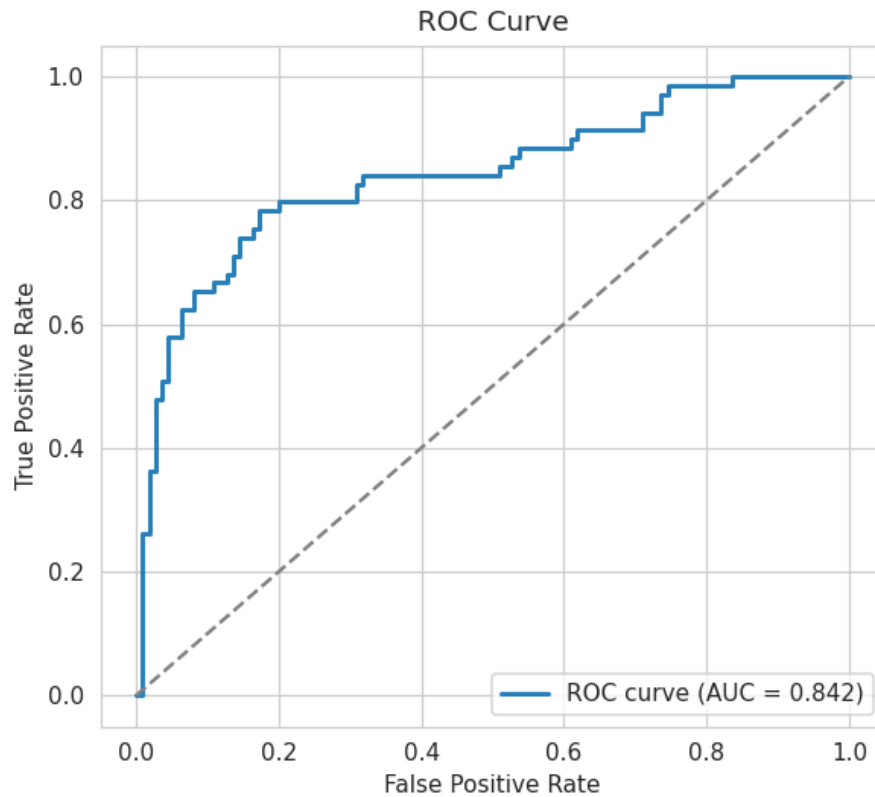


Figure 9. ROC curve.

7.1 Which Metric Matters Most, and Why

For this maritime-safety use case, F1-score (which balances Precision and Recall) is the most important single metric, with Recall given particular weight. The classes are moderately imbalanced (approximately 62/38), so Accuracy alone can be misleading. More importantly, in a safety context the costlier error is a false negative - failing to flag a passenger who is actually at higher survival risk - since that has a higher real-world cost than a false alarm. F1-score, and specifically Recall on the 'Survived' class, better reflect how well the model identifies passengers who need attention.

8. Hyperparameter Tuning

A grid search (5-fold cross-validation, optimizing F1-score) was run over the regularization strength C in $\{0.001, 0.01, 0.1, 1, 10, 100\}$ and penalty type in $\{L1, L2\}$. The best configuration found was `{"C":10,"penalty":"l2","solver":"liblinear"}`, achieving a mean cross-validated F1-score of 0.7268.

Metric	Baseline Model	Tuned Model
Accuracy	0.8045	0.8045
Precision	0.7931	0.7931
Recall	0.6667	0.6667
F1 Score	0.7244	0.7244

In this case, the default Logistic Regression settings were already close to the cross-validated optimum, so tuning produced the same test-set performance while confirming, via grid search, that the chosen regularization strength and penalty type are well-justified rather than arbitrary.

9. Business Insights

1. Which factors most influenced passenger survival?

Sex, Pclass, and Age were the dominant influences, with Fare acting as a correlated proxy for socioeconomic status tied to Pclass.

2. Did passenger class affect survival probability?

Yes, substantially. First-class passengers survived at roughly 63%, second-class at roughly 47%, and third-class at only roughly 24% - consistent with faster lifeboat access from upper decks/classes.

3. Did gender significantly impact survival?

Yes - the single strongest predictor. Female passengers survived at roughly 74% versus roughly 19% for male passengers.

4. What role did age play?

Age had a smaller but meaningful negative coefficient - younger passengers, especially children, had higher survival odds, consistent with the historical 'women and children first' evacuation protocol.

5. Which feature was the strongest predictor?

Sex, based on both its correlation with the target and its logistic regression coefficient magnitude.

6. Recommendations for improving passenger safety in future maritime travel

- Ensure lifeboat capacity and evacuation-route access are equitable across all ticket/cabin classes, not concentrated near upper-class decks.
- Design evacuation drills and signage that reinforce prioritizing children and other vulnerable passengers regardless of cabin class.
- Audit ship layouts for class-based disparities in distance-to-lifeboat and correct structural inequities for lower decks.
- Use lightweight predictive risk-scoring models like this one as decision-support - not sole-decision - tools in emergency-response training and simulation exercises.

10. Conclusion

This project implemented a full machine learning pipeline for Titanic survival prediction using Logistic Regression. After identifying and correcting a significant data-quality issue (418 rows carrying placeholder rather than genuine labels), the model achieved 80.5% accuracy and an F1-score of 0.7244 on held-out data, with Sex, Pclass, and Age emerging as the strongest predictors of survival. Hyperparameter tuning via grid search confirmed the chosen model configuration was near-optimal. Beyond the modeling exercise itself, the analysis reinforces a core lesson of applied data science: careful data understanding and validation are as critical to a trustworthy result as the choice of algorithm.